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## Original Article

### **Influence of illumination conditions on color differences among visual, intraoral scanner, and spectrophotometric tooth shade determination methods: A prospective clinical study**

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## **ABSTRACT**

**Purpose:** In this study, color differences among shade measurements obtained using a visual assessment method, an intraoral scanner, and a spectrophotometric imaging system were evaluated. Additionally, the influence of daylight and fluorescent illumination conditions on intermethod color differences was investigated.

**Materials and Methods:** Shade measurements of maxillary central incisors were performed using three shade determination methods: visual assessment, intraoral scanning, and spectrophotometric imaging-based measurements. These measurements were obtained under standardized daylight and fluorescent lighting conditions. Color differences were calculated using the CIEDE2000 color difference formula. Statistical analyses were performed using nonparametric tests at the 0.05 significance level.

**Results:** Significant differences among shade-determination methods were observed under daylight conditions, whereas none were detected under fluorescent lighting. Illumination conditions significantly influenced color differences associated with visual and intraoral scanner-based measurements. Spectrophotometric imaging-based measurement demonstrated consistent results across illumination environments.

**Conclusions:** Shade-determination outcomes were influenced by both the measurement method and the illumination condition. Visual and intraoral scanner-based measurements exhibited greater sensitivity to lighting variations, whereas spectrophotometric imaging-based measurements remained stable. These findings emphasize the importance of illumination standardization in clinical shade selection procedures.

**Keywords:** Color perception, Dental esthetics, Digital technology, Spectrophotometry, Visual assessment

## Introduction

In contemporary dentistry, particularly in anterior restorations, accurate shade determination is essential for achieving esthetic outcomes and patient satisfaction, making tooth shade selection a challenging task for dental practitioners<sup>1</sup>. Tooth color can be determined using subjective or objective methods, and although visual shade selection with commercially available shade guides is the most practical and cost-effective approach in daily practice, it remains inherently subjective and prone to variability<sup>2,3</sup>. This method relies on direct comparison of the target tooth with adjacent teeth for rapid chairside decision-making, but its reliability is influenced by environmental and observer-related factors such as lighting conditions, viewing distance, oral environment, color perception, clinical experience, and eye fatigue<sup>4-10</sup>.

Owing to the limitations of visual shade selection, objective and rapid color-measuring devices such as spectrophotometers, spectroradiometers, and colorimeters have been developed<sup>11-13</sup>. Advances in digital workflows have also led to the widespread use of digital cameras, intraoral scanners, and mobile phones for color analysis<sup>14-16</sup>.

Owing to their high accuracy and reproducibility, spectrophotometers are widely accepted as reliable tools for tooth color determination<sup>17-19</sup>. They analyze the full spectrum and radiant intensity of reflected light using spectral filters dividing visible light into 1-25 nm intervals<sup>20</sup>, generating CIE L\*, a\*, and b\* values that can be converted into shade systems. However, their performance may be affected by ambient lighting, device positioning, surface texture, and the natural translucency and gloss of teeth<sup>19</sup>.

With advances in digital technology, clinicians can now work in a virtual environment that enhances diagnosis, planning, and treatment<sup>21</sup>. Intraoral scanners enable digital data acquisition through point-by-point scanning using light or laser technology, and their ability to simultaneously record geometric and color information offers a clinical advantage that has increased research interest in their use for shade determination. Although some studies show intraoral scanners yielding more consistent results than visual assessment, their color measurement performance is influenced by algorithm structure, sensor type, and lighting conditions<sup>22-24</sup>.

Previous studies on tooth color determination have mainly focused on comparisons between visual and spectrophotometric or digital methods and on the performance of digital systems such as intraoral scanners. Most report higher accuracy and reproducibility for spectrophotometric systems than for visual assessment<sup>3,6,17-20</sup>, whereas digital systems provide more consistent results but remain influenced by sensor technology, software algorithms, and environmental conditions<sup>14,15,21,22</sup>.

Several studies have shown that illumination conditions significantly influence color perception and measurement outcomes in visual and camera-based systems<sup>4,7,13</sup>, and that differences in the spectral distribution of light sources may alter measured CIELAB values<sup>20</sup>. However, most available evidence is limited to in vitro studies or comparisons between only two methods<sup>3,6,14,15,22</sup>, and comprehensive in vivo studies simultaneously evaluating spectrophotometric imaging, intraoral scanners, and visual assessment under different illumination conditions remain scarce. The present study therefore evaluated these three shade-determination methods under daylight and fluorescent illumination in vivo, providing information on the combined effects of measurement method and illumination environment on clinically relevant color differences ( $\Delta E_{00}$ ).

The aim of this prospective in vivo study was to evaluate  $\Delta E_{00}$  color differences among shade measurements obtained via visual assessment, an intraoral scanner, and a spectrophotometric imaging system, and to investigate the influence of daylight and fluorescent lighting on intermethod color differences, comparing measurements from the same tooth under standardized illumination and measurement conditions. Two null hypotheses were formulated: no significant differences in  $\Delta E_{00}$  values would exist among the shade-determination methods, and illumination conditions would not significantly influence intermethod  $\Delta E_{00}$  values.

## Materials and Methods

This prospective clinical study protocol was approved by the university's ethics committee (Approval No: D-KA25/41; Date: 26.11.2025) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants before enrollment.

## Sample size estimation

Sample size was estimated using G\*Power (version 3.1.9.2) for a within-subject repeated-measures design with three shade-determination methods and two illumination conditions. Assuming an effect size of  $f = 0.56$ , based on previously reported  $\Delta E_{00}$  values for spectrophotometric and digital shade-selection systems<sup>25</sup>,  $\alpha = 0.05$ , and 95% power, the minimum required sample size was 46 participants. ANOVA-based estimation was selected during planning, when the data distribution was still unknown; nonparametric tests were subsequently used for the final analyses because normality assumptions were violated. To compensate for potential data loss, 48 participants were included.

#### **Inclusion criteria:**

- Females/males between 20 and 40 years of age,
- Absence of a restoration, crack, discoloration, or structural defect in the upper central incisor,
- Being periodontally healthy, defined as the absence of clinical attachment loss, probing pocket depths  $\leq 3$  mm, and no signs of gingival inflammation or bleeding on probing.
- Not having received teeth whitening treatment during the past 12 months.

#### **Exclusion criteria:**

- History of trauma or restoration in the anterior region,
- Conditions that may cause intraoral or systemic discoloration,
- Heavy smoking,
- Pregnancy,
- Presence of clinical conditions that may prevent compliance with the study protocol.

A total of 48 participants (33 females, 15 males; age range 20-40 years) were included, and color analysis of tooth #11 was performed for each participant using a shade guide system (VITA

3D-MASTER), an intraoral scanner (3Shape TRIOS 3), and a spectrophotometric imaging device (SpectroShade Micro) under both natural daylight and fluorescent lighting.

All color measurements were performed in accordance with the recommendations of ISO/TR 28642:2016 (Dentistry-Guidance on color measurement), to ensure standardization of observer conditions, illumination, and measurement procedures.

### **Standardization of illumination and measurement conditions**

Color analysis of tooth #11 was performed under two lighting conditions for each participant: natural daylight ( $\approx 5500$ - $6500$  K) and standardized fluorescent illumination ( $6500$  K; Osram 49 W/865). The correlated color temperature of the fluorescent source was comparable to that of CIE standard illuminant D65; however, spectral equivalence was not assumed, and all measurements were conducted under controlled, consistent lighting at the tooth surface.

Daylight measurements were performed between 09:00 and 11:00, in the same clinical room, to reduce variability associated with changes in natural daylight throughout the day; measurements were avoided during rapidly changing weather conditions whenever possible.

All measurements were performed against a neutral gray background to minimize chromatic adaptation and contrast effects, with patients positioned upright and shade assessed at a standardized viewing distance (20-30 cm) and angle. Participants removed lipstick, makeup, or other products that could affect measurements, and the tooth surface was cleaned with water spray and gently air-dried, avoiding prolonged drying or enamel dehydration.

### **Visual shade selection**

Visual shade selection was performed by a single female prosthodontist with normal color vision, verified by the Farnsworth-Munsell 100 Hue Test, using the VITA 3D-MASTER shade guide (VITA Zahnfabrik, Bad Säckingen, Germany) (Fig 1A). To assess intra-observer reliability,

repeated visual shade selections were performed in 10 randomly selected participants at a 2-week interval by the same examiner under standardized conditions; re-evaluating the entire sample was avoided to limit participant burden and clinical workload. Intra-class correlation coefficients (ICC) were calculated to evaluate the consistency and reproducibility of these repeated assessments.

Evaluation time was limited to 10 seconds to minimize visual fatigue, with the shade guide held parallel to the tooth surface and the clinician's line of sight perpendicular to it (Fig 1B). Value was selected first, followed by chroma within the selected column and then hue. Separate shade records were obtained under both lighting conditions, with each visual assessment performed three times per condition.

### **Shade selection using an intraoral scanner**

Digital shade measurement was performed using a 3Shape TRIOS 3 intraoral scanner (3Shape, Copenhagen, Denmark) with TRIOS software version 25.1 (Fig. 2A), by the same operator under identical clinical conditions. White balance calibration, and calibration of the scanner's operating system and shade-determination module, were performed prior to measurements. Scans were conducted under both illumination conditions, covering the maxillary anterior region from the right to left canine (FDI 13-23) to standardize scan volume across participants. A standardized scanning path was applied: incisal surfaces continuously from right to left canine, then palatal surfaces left-to-right, then labial surfaces right-to-left. To reduce operator-dependent variability, the number of image frames captured per scan was maintained at approximately 1200-1500. After recording, the principal vestibular area was marked and the VITA 3D-MASTER shade displayed (Fig 2B). Three consecutive scans were performed per illumination condition, and consistency was confirmed before recording the final measurement.

### **Shade selection using a spectrophotometric imaging system**



Objective color measurements were performed using a spectrophotometric imaging device (SpectroShade, MHT Optic Research, Niederhasli, Switzerland) by a single experienced operator (Figure 3A), calibrated per manufacturer instructions before each session. Measurements followed the system's standardized acquisition protocol, with the head positioned perpendicular to the tooth surface at a consistent working distance. Under each lighting condition, three consecutive image-based measurements were obtained for tooth #11, the measurement area was standardized to the middle third of the labial surface using the device software's tooth-mapping interface, and the L, a\*, and b\* values provided by the device were recorded (Fig 3B-C).

### Color difference ( $\Delta E_{00}$ ) calculations

Color differences ( $\Delta E_{00}$ ) among the three shade-determination methods (visual assessment, intraoral scanning, and spectrophotometric imaging) were calculated using the CIEDE2000 formula, which better reflects perceptual color differences. CIELAB ( $L^*$ ,  $a^*$ ,  $b^*$ ) coordinates were obtained directly from spectrophotometric imaging measurements, whereas for visual shade selection and intraoral scanner measurements the VITA 3D-MASTER shade codes were converted into corresponding CIELAB coordinates using the conversion table reported by Budde et al<sup>24</sup>, generated from spectrophotometrically derived CIELAB reference coordinates for all 29 VITA 3D-MASTER shade tabs. For pairwise comparisons,  $\Delta E_{00}$ -VS represented the difference between visual assessment and spectrophotometric imaging,  $\Delta E_{00}$ -ST between spectrophotometric imaging and intraoral scanning, and  $\Delta E_{00}$ -VT between visual assessment and intraoral scanning.

The CIEDE2000 ( $\Delta E_{00}$ ) formula is as follows:

$$\Delta E_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \left(\frac{\Delta C'}{k_C S_C}\right) \left(\frac{\Delta H'}{k_H S_H}\right)}$$

For pairwise comparisons, the measurements from the two compared methods were designated L1, a1, b1 and L2, a2, b2, respectively. The acquired L, a\*, and b\* values were entered into a CIEDE2000 formula prepared in an Excel spreadsheet, with L\* representing lightness and a\*/b\* representing the red-green and yellow-blue chromatic axes (dimensionless values). Weighting factors were set as  $kL = 2$ ,  $kC = 1$ , and  $kH = 1$ , consistent with previous dental color-difference studies<sup>25,26</sup>.

$\Delta E_{00}$  values were interpreted according to previously established perceptibility ( $\approx 0.8$ ) and acceptability ( $\approx 1.8$ ) thresholds<sup>23</sup>.

For each participant, three repeated measurements were obtained per method under each illumination condition; to minimize random variability, the mean of the three was used for all analyses, and repeated measurements were not entered as independent observations.

### **Statistical analysis**

Statistical analyses were performed using IBM SPSS Statistics (version 20.0; Chicago, IL, USA), with  $\Delta E_{00}$  as the primary outcome. Descriptive statistics were expressed as mean  $\pm$  SD and median (min-max). Data distribution was assessed with the Shapiro-Wilk test; because  $\Delta E_{00}$  values were not normally distributed, nonparametric methods were applied. Within-subject comparisons between daylight and fluorescent conditions were performed with the Wilcoxon signed-rank test, and comparisons among the three methods under the same illumination were performed with the Friedman test, followed by pairwise post hoc comparisons with adjusted significance levels when appropriate. Significance was set at  $\alpha = 0.05$ .

### **Results**

The intra-observer reliability of visual shade selection was good (ICC = 0.89).

Three shade-determination methods were applied to each participant under both lighting conditions. A statistically significant difference was observed among methods under daylight ( $p < 0.01$ ), with Friedman post hoc analysis showing  $\Delta E_{00}\text{-ST}$  significantly greater than both  $\Delta E_{00}\text{-VS}$  and  $\Delta E_{00}\text{-VT}$  ( $p < 0.05$ ); no difference was found between  $\Delta E_{00}\text{-VS}$  and  $\Delta E_{00}\text{-VT}$ . Under fluorescent lighting, no significant differences were found among  $\Delta E_{00}\text{-VS}$ ,  $\Delta E_{00}\text{-ST}$ , and  $\Delta E_{00}\text{-VT}$  ( $p > 0.05$ ; Table 1).

By lighting condition,  $\Delta E_{00}\text{-VS}$  and  $\Delta E_{00}\text{-VT}$  values were significantly greater under fluorescent than daylight lighting ( $p < 0.05$  for both), whereas  $\Delta E_{00}\text{-ST}$  did not differ significantly between conditions ( $p > 0.05$ ; Table 2).

## Discussion

Based on the findings of this study, the first null hypothesis was partially rejected. Statistically significant differences among shade-determination methods were observed under daylight conditions, whereas no significant differences were detected under fluorescent lighting.

The statistically significant intermethod differences observed under daylight are consistent with previous reports that different shade-matching techniques can yield non-equivalent results even under controlled conditions. Kröger et al<sup>4</sup> found that both instrumental and visual shade-determination methods varied across illuminants, and Hardan et al<sup>11</sup>, in a systematic review and meta-analysis, emphasized that visual shade selection is highly susceptible to lighting conditions and less reliable than instrumental methods. Regarding intraoral scanners, Tabatabaian et al<sup>14</sup> showed that accuracy and precision are inconsistent and vary with sensor characteristics, acquisition protocols, and software algorithms, a variability also reported by Huang et al<sup>15</sup>, who attributed discrepancies to differences in optical systems and data processing. In contrast, spectrophotometric systems have been reported to provide more reproducible measurements;

Rutkūnas et al<sup>21</sup> highlighted their greater consistency in tooth color determination. Together, these findings suggest that both methodological differences between systems and the sensitivity of visual assessment to illumination contributed to the discrepancies observed under daylight conditions in the present study.

The findings for visual assessment further highlight the influence of illumination on color perception, which is affected not only by tooth optical properties but also by the spectral distribution of the light source, observer adaptation, and contrast effects a phenomenon associated with metamerism and chromatic adaptation<sup>4,11</sup>. The  $\Delta E_{00}$  variability observed here may therefore reflect perceptual shifts rather than true color differences; because visual shade selection showed good intra-observer reliability (ICC = 0.89), this variability is more likely attributable to illumination and perceptual factors than to examiner inconsistency.

The second null hypothesis was partially rejected, as illumination affected visual and intraoral scanner-based measurements but not spectrophotometric measurements. Although daylight's broad spectral distribution and dynamic nature may introduce variability, previous studies similarly report that both visual and instrumental shade determination can be influenced by the illuminant used<sup>4</sup>, with scanner-related factors contributing further<sup>14,15</sup>, while instrumental systems remain more consistent<sup>21</sup>.

An important observation was that the significant intermethod differences seen under daylight disappeared under fluorescent illumination which should not be interpreted as evidence of similar method performance. Natural daylight has a broad, continuous spectral power distribution and high color rendering index (CRI), allowing subtle color and device-specific differences to be detected more readily<sup>4,20</sup>, whereas fluorescent light typically has a discontinuous spectrum concentrated at specific wavelengths, which may reduce the ability of both observers

and optical systems to discriminate fine color differences<sup>4</sup>. The absence of significant differences under fluorescent lighting may therefore reflect reduced sensitivity of the lighting environment rather than true convergence among methods.

Visual and intraoral scanner-based measurements showed greater variability across illumination conditions, whereas spectrophotometric measurements remained unaffected, consistent with reports that spectrophotometric systems provide higher reliability due to standardized spectral measurement principles<sup>20</sup>. Budde et al<sup>24</sup> similarly reported high validity and reliability for spectrophotometric systems, noting that area-based systems such as the imaging device used here reduce the edge-loss and localized variation seen with point-based systems, which may explain the stability observed in this study.

Differences in device architecture may also influence susceptibility to ambient illumination: spectrophotometric imaging devices use standardized acquisition geometries and spectral protocols designed specifically for color measurement<sup>24</sup>, whereas intraoral scanners, optimized primarily for surface capture, rely on device-specific color-reconstruction algorithms that may be more sensitive to ambient light. Although not directly investigated, this difference may partly explain why imaging measurements were less affected by illumination than scanner-based measurements.

The clinical interpretation of  $\Delta E_{00}$  values is particularly important for understanding the practical implications of these findings. Reported perceptibility and acceptability thresholds are approximately 0.8 and 1.8, respectively<sup>23</sup>. Most  $\Delta E_{00}$  values observed here exceeded the perceptibility threshold, suggesting intermethod color differences may be visually detectable; however, because some values remained within acceptability limits, these differences may not necessarily be clinically unacceptable<sup>20</sup>. Nevertheless, several mean  $\Delta E_{00}$  values exceeded the

commonly accepted acceptability threshold, indicating that shade determinations from different methods may not always be clinically interchangeable, particularly under certain illumination conditions.

An important methodological consideration concerns the derivation of the CIELAB coordinates used for  $\Delta E_{00}$  calculations. While the spectrophotometric imaging system provided direct  $L^*$ ,  $a^*$ , and  $b^*$  values, visual and intraoral scanner measurements were converted from VITA 3D-MASTER shade outputs using reference tables; part of the observed  $\Delta E_{00}$  differences may therefore reflect uncertainties in shade-to-CIELAB conversion rather than true differences in color-acquisition performance. Because shade tabs represent discrete color categories rather than continuous coordinates, this conversion may reduce color resolution and introduce additional variability. Comparisons involving visual and intraoral scanner-based measurements should therefore be interpreted with caution, and future studies should prioritize systems capable of providing direct CIELAB outputs.

### Limitations

The findings of this study should be interpreted in light of several limitations:

- The evaluation was restricted to a single tooth type and a limited age range, which may affect the generalizability of the findings.
- Only one intraoral scanner and one spectrophotometric imaging system were evaluated; therefore, the results may not be directly applicable to other devices with different optical configurations or software algorithms.
- Unlike the spectrophotometric imaging system, which directly provides CIE  $L^*$ ,  $a^*$ , and  $b^*$  coordinates, the intraoral scanner and visual shade selection methods generated shade guide-based outputs requiring conversion into CIELAB coordinates via reference tables. Although used

previously, this approach does not represent direct instrumental measurement and may have introduced additional uncertainty into  $\Delta E_{00}$  calculations; the reported color differences should therefore be interpreted as estimates combining direct measurements and shade-based approximations.

- Although a standardized scanning protocol was applied, intraoral scanning inherently relies on image-stitching procedures; cumulative stitching errors, scan-volume-related distortions, or minor acquisition variations may therefore have influenced the recorded shade information and derived color coordinates.
- Although illumination conditions were standardized and daylight measurements were performed during fixed morning hours, illuminance (lux), color rendering index (CRI), and spectral power distribution (SPD) were not objectively measured. Therefore, some variability in the daylight environment cannot be completely excluded.
- Consequently, the proposed explanation that fluorescent illumination may have reduced the sensitivity of the measurement environment to reveal intermethod differences should be considered interpretative rather than directly verified by the present experimental design.

## Conclusion

Within the limitations of this current study, it was concluded that:

1. Color differences among shade-determination methods differed under daylight conditions but not under fluorescent illumination.
2. Illumination influenced color differences for visual assessment and intraoral scanner measurements but not for spectrophotometric imaging measurements.

3. Although spectrophotometric imaging-based measurements appeared less affected by illumination changes, these findings do not allow definitive conclusions regarding the superiority of one shade-determination method over another.

## **Declarations**

## **Abbreviations**

$\Delta E_{00}$ : CIEDE2000 color difference

CIELAB: Commission Internationale de l'Eclairage L\*, a\*, b\* color space

CIE: Commission Internationale de l'Eclairage

ICC: Intraclass Correlation Coefficient

ANOVA: Analysis of Variance

ISO: International Organization for Standardization

FDI: Federation Dentaire Internationale

CRI: Color Rendering Index

SPD: Spectral Power Distribution

## **Ethics approval and consent to participate**

This prospective clinical study protocol was approved by the Başkent University's ethics committee (Approval No: D-KA25/41; Date: 26.11.2025).

In this study, all procedures were carried out in accordance with ethical principles and the Declaration of Helsinki. Written and verbally informed consent was obtained from the volunteers after they were informed about the study and agreed to participate.

**Clinical trial number:** Not applicable.

**Consent for publication:** Not applicable.

**Availability of data and materials**



The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

### **Competing interests**

The authors declare no competing interests.

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No funding was received for this study.

### **Author Contributions**

Ecem Sancar: Conceptualization, supervision, Manuscript writing.

Elif Kocak: Data collection, methodology design, statistical analysis, interpretation of results supervision, manuscript writing.

Sezin Deniz Gurbuz: Data collection, methodology design, statistical analysis, interpretation of results supervision, manuscript writing.

Melis Topaloglu: Data collection, methodology design, statistical analysis, interpretation of results supervision, manuscript writing.

Mert Anıl Yıldız: Data collection, methodology design, statistical analysis, interpretation of results supervision, manuscript writing.

Selim Erkut: Project administration, supervision, and final approval of the manuscript.

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**Table 1.** Comparison of intermethod  $\Delta E_{00}$  values under daylight and fluorescent lighting conditions

| <b>Lighting Condition</b> | <b><math>\Delta E_{00}</math> -VS</b> | <b><math>\Delta E_{00}</math> -ST</b> | <b><math>\Delta E_{00}</math> -VT</b> |         |
|---------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------|
|                           | Mean±SD                               | Mean±SD                               | Mean±SD                               | p value |
|                           | Median (Min-Max)                      | Median (Min-Max)                      | Median (Min-Max)                      |         |

|                    |             |             |             |                              |
|--------------------|-------------|-------------|-------------|------------------------------|
| <b>Daylight</b>    | 2.1±1.7     | 2.9±1.7     | 2.1±1.9     | <b>&lt;0.001<sup>f</sup></b> |
|                    | 2.4 (0-6.8) | 2.9 (0-6.5) | 2.7 (0-9.0) |                              |
| <b>Fluorescent</b> | 3.0±1.9     | 2.8±2.1     | 3.1±2.4     | 0.885 <sup>f</sup>           |
| <b>light</b>       | 3.4 (0-7.8) | 3.4 (0-7.8) | 3.4 (0-8.9) |                              |

f: Friedman test

$\Delta E_{00}$ -VS: Color difference between visual and spectrophotometric imaging-based measurements.

$\Delta E_{00}$ -ST: Color difference between spectrophotometric imaging and intraoral scanner-based measurements.

$\Delta E_{00}$ -VT: Color difference between visual and intraoral scanner-based measurements.

**Table 2.** Comparison of  $\Delta E_{00}$  values between shade determination methods according to lighting conditions

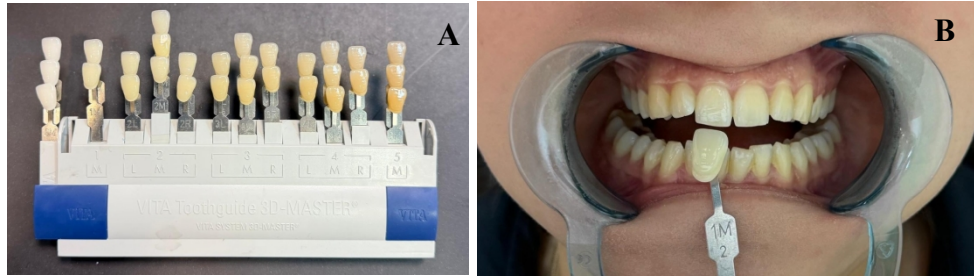
|                                       | <b>Daylight</b>  | <b>Fluorescent light</b> |                          |
|---------------------------------------|------------------|--------------------------|--------------------------|
|                                       | Mean±SD          | Mean±SD                  | P value                  |
|                                       | Median (Min–Max) | Median (Min–Max)         |                          |
| <b><math>\Delta E_{00}</math> -VS</b> | 2.1±1.7          | 3.0±1.9                  | <b>0.025<sup>b</sup></b> |
|                                       | 2.4 (0–6.8)      | 3.4 (0–7.8)              |                          |
| <b><math>\Delta E_{00}</math> -ST</b> | 2.9±1.7          | 2.8±2.1                  | 0.674 <sup>b</sup>       |
|                                       | 2.9 (0–6.5)      | 3.4 (0–7.8)              |                          |
| <b><math>\Delta E_{00}</math> -VT</b> | 2.1±1.9          | 3.1±2.4                  | <b>0.017<sup>b</sup></b> |
|                                       | 2.7 (0–9.0)      | 3.4 (0–8.9)              |                          |

b: Wilcoxon test

$\Delta E_{00}$ -VS: Color difference between visual and spectrophotometric imaging-based measurements.

$\Delta E_{00}$ -ST: Color difference between spectrophotometric imaging and intraoral scanner-based measurements.

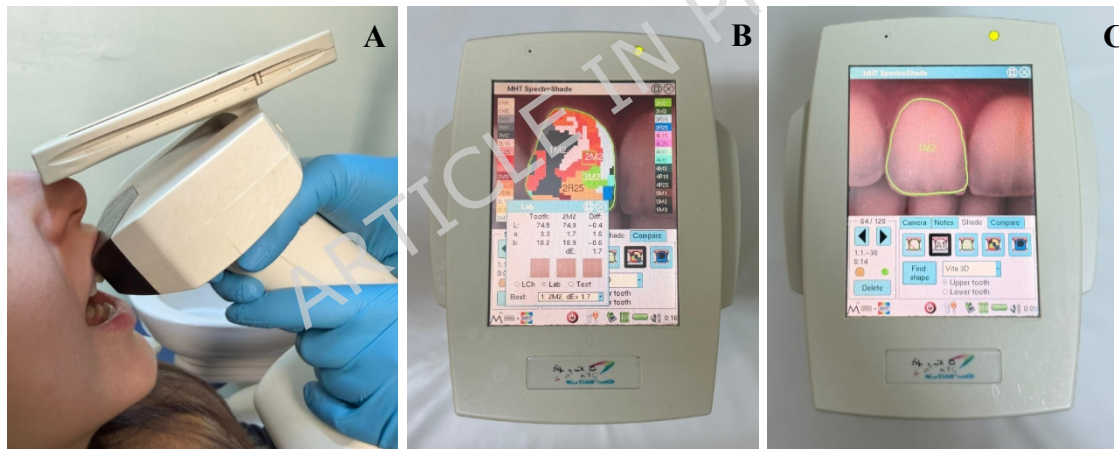
$\Delta E_{00}$ -VT: Color difference between visual and intraoral scanner-based measurements.



**Figure 1.** (A) VITA 3D-MASTER shade guide system and (B) visual color selection with a shade guide.



**Figure 2.** (A) The intraoral scanner tested in this study (3Shape TRIOS 3, Copenhagen, Denmark). (B) Screenshot of the digital color matching on the right central incisor according to the VITA 3D-MASTER shade guide system.



**Figure 3.** (A) Spectrophotometric imaging-based shade analysis using the SpectroShade system (SpectroShade, MHT Optic Research, Niederhasli, Switzerland); (B–C) Output of the tooth shade in the VITA 3D-MASTER system.